

Al al-Bayt University

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Department of Information Systems

Student Performance Prediction Using Data Mining Techniques

Final Project Report

Course	Data Mining - 902432
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Abstract

This project applies data mining and machine learning techniques to predict whether a student will pass or fail based on academic, demographic, social, and school-related variables. The project uses the UCI Student Performance dataset, specifically the Portuguese course dataset. The final grade G3 is converted into a binary target variable: Pass if G3 is greater than or equal to 10, and Fail otherwise. The work includes data understanding, preprocessing, exploratory data analysis, model implementation, evaluation, and interpretation. Four supervised classification algorithms were compared: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine. The best overall result was achieved by Logistic Regression when previous grades G1 and G2 were included, with an F1-score of 0.9455. A second experiment excluding G1 and G2 was also conducted to test a more realistic early-warning scenario.

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1. Introduction

Data mining is the process of discovering useful patterns, relationships, and trends from data. In educational environments, data mining can support decision-making by identifying students who may need additional academic support before final outcomes occur.

The selected project topic is student performance prediction. The project goal is to build a predictive model that classifies students into Pass or Fail categories using available student attributes. This topic is relevant because it supports early intervention, academic advising, and evidence-based educational planning.

The project was implemented using Python on Kaggle, with pandas for data handling, matplotlib and seaborn for visualization, and scikit-learn for machine learning pipelines and evaluation.

2. Problem Definition and Objectives

Problem Statement. Given a set of student-related attributes, can a supervised machine learning model predict whether a student will pass or fail the final course outcome?

- Understand and describe the student performance dataset.
- Clean and preprocess numerical and categorical attributes.
- Visualize key relationships between academic outcomes and student factors.
- Build multiple classification models and compare their performance.
- Interpret the best model and identify important predictive features.
- Discuss practical insights and limitations for educational decision-making.

3. Dataset Description

The dataset used in this project is the Student Performance dataset. The project used the Portuguese course file, student-por.csv. The cleaned dataset contains 649 student records and 33 original attributes. After target engineering, the dataset contains 34 columns including pass_final.

The attributes include school, sex, age, address type, family size, parents' education, parents' jobs, travel time, study time, previous failures, school support, family support, extracurricular activities, higher education intention, internet access, family relationship quality, free time, going out, alcohol consumption, health, absences, and course grades G1, G2, and G3.

The original output variable G3 represents the final grade on a 0 to 20 scale. For this classification project, G3 was transformed into a binary target: Pass = 1 when $G3 \geq 10$, and Fail = 0 when $G3 < 10$.

Dataset Property	Value
Dataset file	student-por.csv
Number of records	649
Original attributes	33
Engineered target	pass_final
Prediction type	Binary classification
Pass rule	$G3 \geq 10$
Fail rule	$G3 < 10$
Missing values detected	0
Duplicate rows detected after cleaning	0

4. Data Preprocessing

- Dataset loading: the CSV file was loaded using pandas with semicolon as the separator.
- Data inspection: shape, column names, data types, descriptive statistics, missing values, and duplicated rows were checked.
- Target engineering: pass_final was created from G3 using the threshold $G3 \geq 10$.
- Feature separation: G3 and pass_final were removed from the input features to prevent direct leakage from the target.
- Categorical encoding: categorical variables were encoded using OneHotEncoder with handle_unknown set to ignore.
- Numerical preprocessing: numerical variables were imputed using the median and standardized using StandardScaler.
- Train/test split: the dataset was divided into 80% training and 20% testing with stratification to preserve the class distribution.
- Two experiments were designed: one including previous grades G1 and G2, and one excluding them to simulate earlier prediction without prior grades.

Preprocessing Pipeline. The implementation used a scikit-learn ColumnTransformer and Pipeline structure. This keeps preprocessing and model training connected, reduces manual errors, and makes the experiment reproducible.

5. Exploratory Data Analysis

This section summarizes the main visual findings from the dataset.

Figure 1. Target distribution

Most students passed the course: 549 students passed and 100 students failed. This indicates a class imbalance where the Pass class represents approximately 84.6% of the dataset.

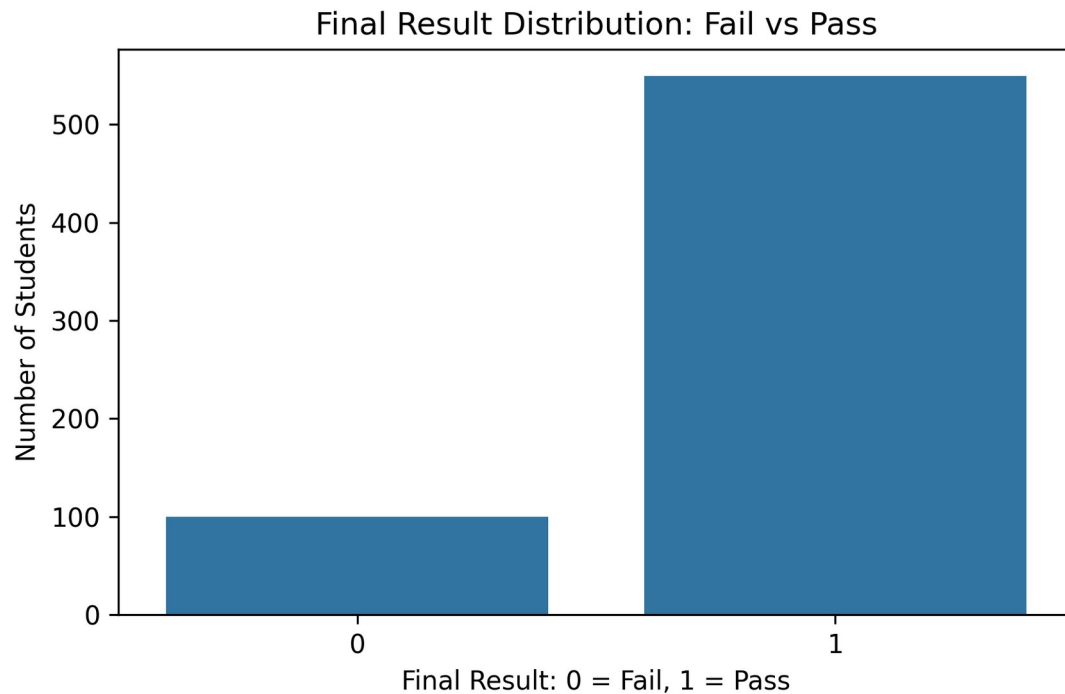


Figure 2. Final grade distribution

The final grade G3 is concentrated around grades 10 to 14, with a few very low scores. The mean final grade is approximately 11.91, and the maximum is 19.

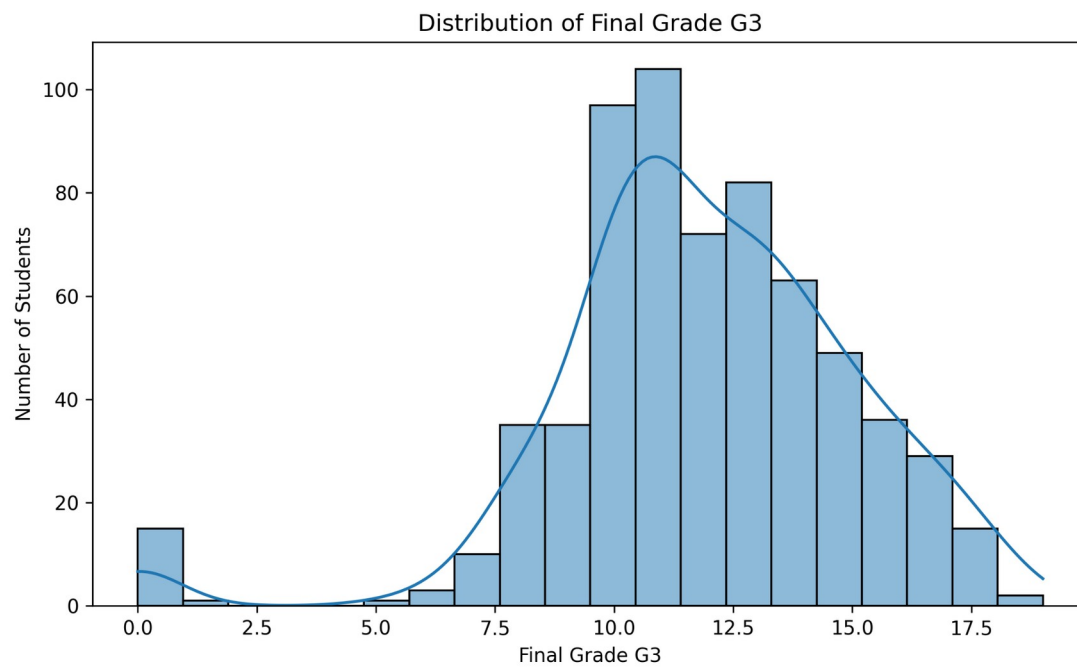
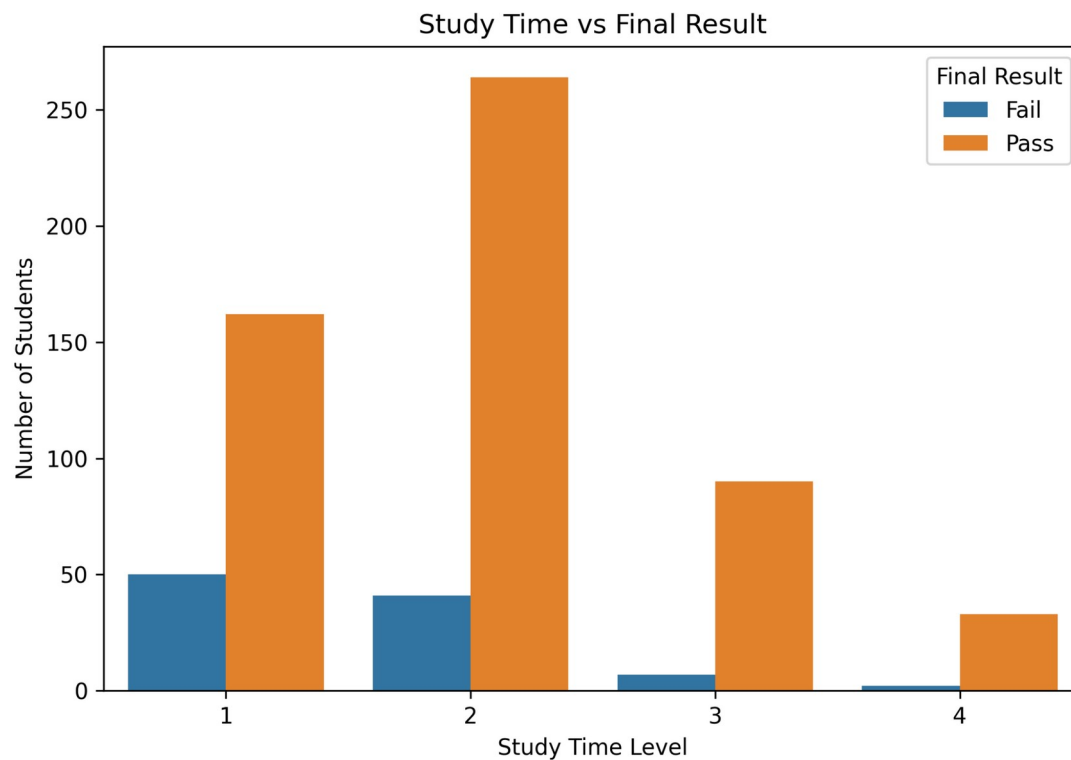


Figure 3. Study time vs final result

Study time level 2 has the largest number of passing students. Higher study time levels show fewer total students, but the fail count is small among levels 3 and 4.

**Figure 4. Previous failures vs final result**

Students with zero previous failures have a much higher pass count. As the number of previous failures increases, the proportion of failing students becomes more important. This suggests that previous failures are a strong risk factor.

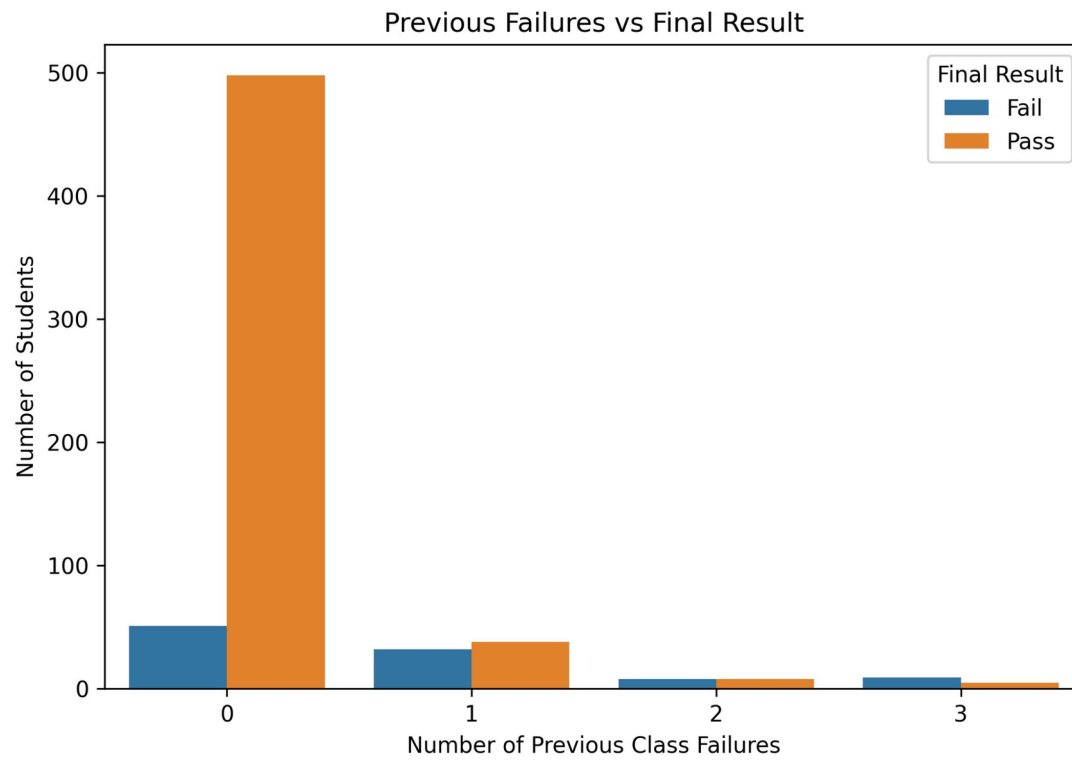
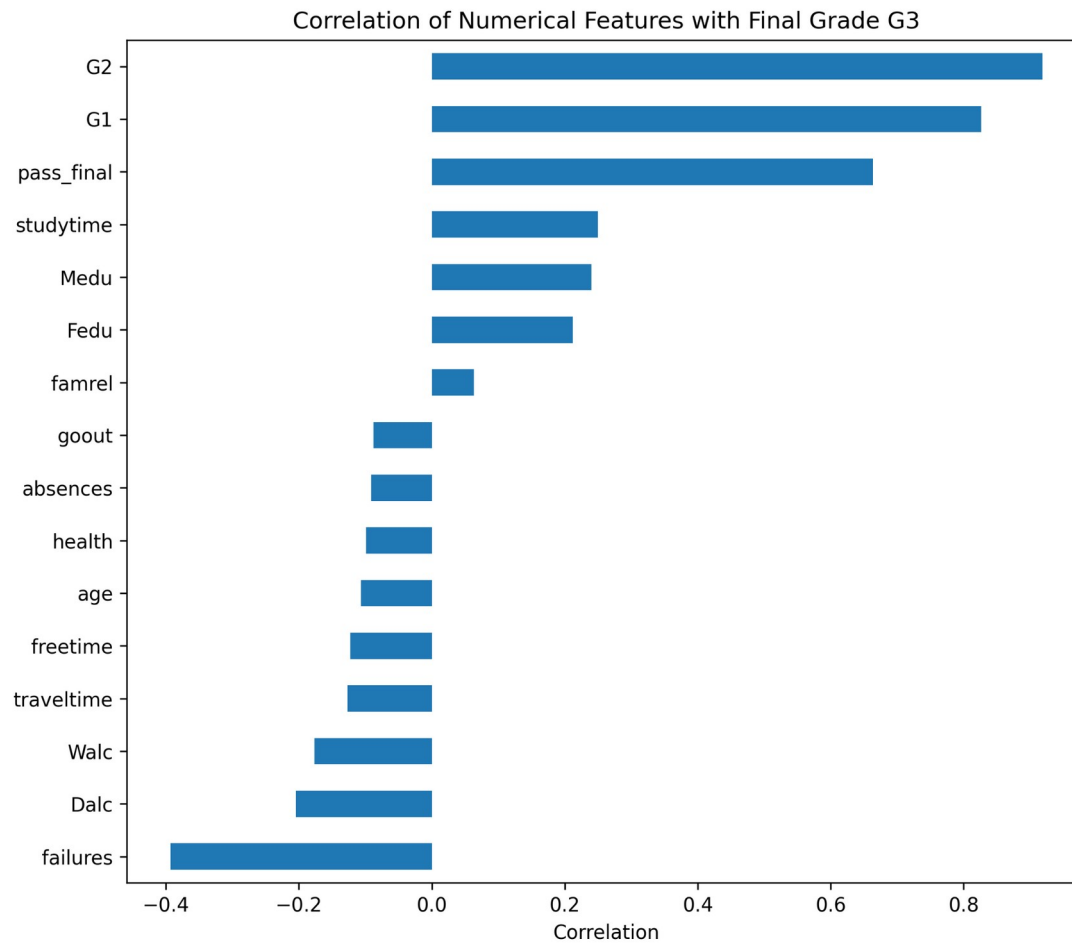


Figure 5. Correlation of numerical features with G3

The strongest positive correlations with the final grade are G2 and G1. Study time, mother education, and father education have smaller positive correlations. Previous failures show the strongest negative correlation with G3.



6. Methodology and Model Implementation

The project used supervised binary classification. Four models were trained and evaluated using the same preprocessing pipeline and the same train/test split to ensure a fair comparison.

The first experiment included previous grades G1 and G2 as input features. This experiment estimates final performance when earlier period grades are already known.

The second experiment excluded G1 and G2. This experiment is more realistic for early warning because it tries to identify at-risk students using demographic, social, behavioral, and support-related factors before grade information is available.

Models Used

Model	Purpose in This Project
Logistic Regression	A linear classification model that estimates the probability of passing.
Decision Tree	A rule-based model that splits data according to feature conditions.
Random Forest	An ensemble of decision trees that improves stability and reduces overfitting.
Support Vector Machine	A margin-based classifier using an RBF kernel for nonlinear separation.

Evaluation Metrics

Accuracy: The overall proportion of correct predictions.

Precision: Among predicted Pass cases, how many were actually Pass.

Recall: Among actual Pass cases, how many were correctly detected.

F1-score: The harmonic mean of precision and recall; useful with imbalanced classes.

ROC-AUC: Measures ranking ability across classification thresholds.

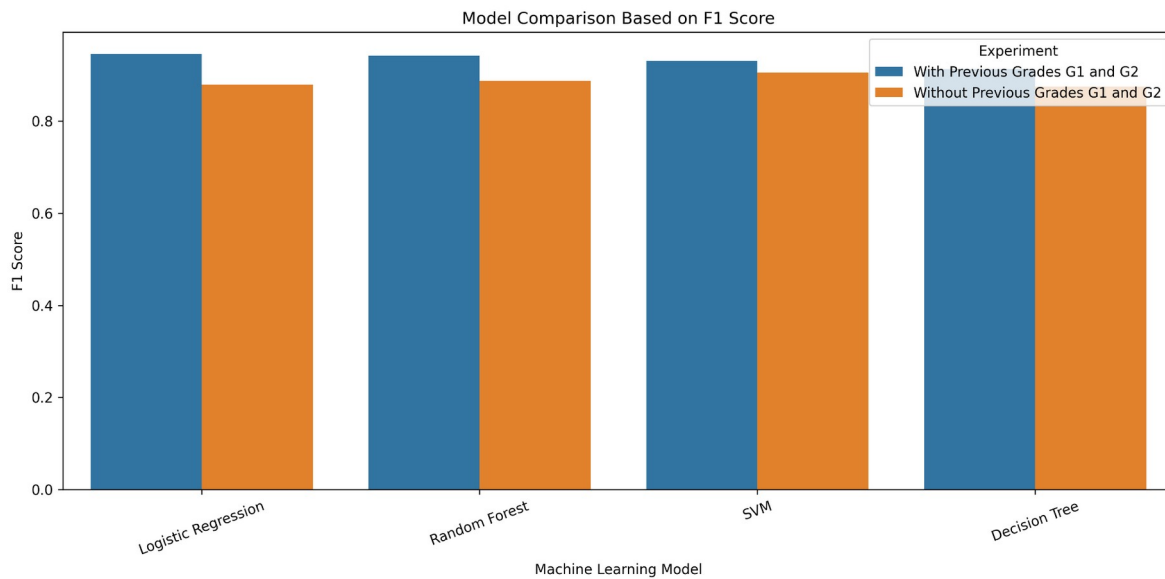
7. Model Evaluation Results

The following table shows the final comparison of the evaluated models.

Experiment	Model	Accuracy	Precision	Recall	F1	ROC_AUC
With Previous Grades G1 and G2	Logistic Regression	0.9077	0.9455	0.9455	0.9455	0.9273
With Previous Grades G1 and G2	Random Forest	0.9000	0.9292	0.9545	0.9417	0.9309
With Previous Grades G1 and G2	SVM	0.8769	0.8983	0.9636	0.9298	0.9168
With Previous Grades G1 and G2	Decision Tree	0.8538	0.9174	0.9091	0.9132	0.7295
Without Previous Grades G1 and G2	SVM	0.8308	0.8607	0.9545	0.9052	0.6691
Without Previous Grades G1 and G2	Random Forest	0.8000	0.8500	0.9273	0.8870	0.6557
Without Previous Grades G1 and G2	Logistic Regression	0.7923	0.8673	0.8909	0.8789	0.6227
Without Previous Grades G1 and G2	Decision Tree	0.7846	0.8596	0.8909	0.8750	0.5455

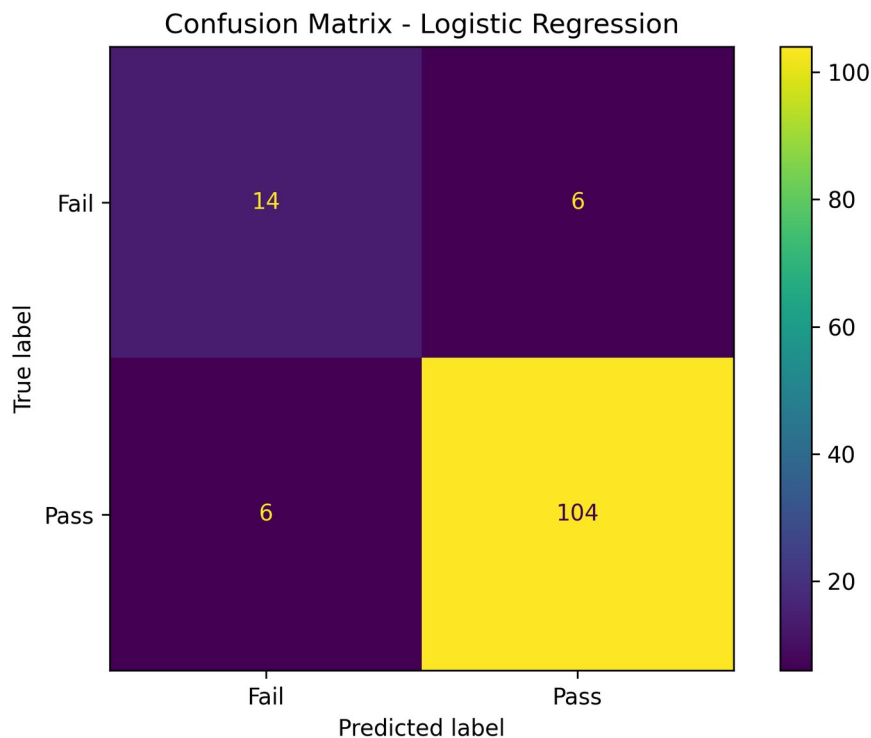
Figure 6. Model comparison based on F1-score

The best model was Logistic Regression in the experiment with G1 and G2, achieving $F1 = 0.9455$. Random Forest was very close with $F1 = 0.9417$. Without previous grades, SVM achieved the highest F1-score among the early-warning models, with $F1 = 0.9052$.



8. Interpretation and Discussion

Figure 7. Confusion matrix of the best model



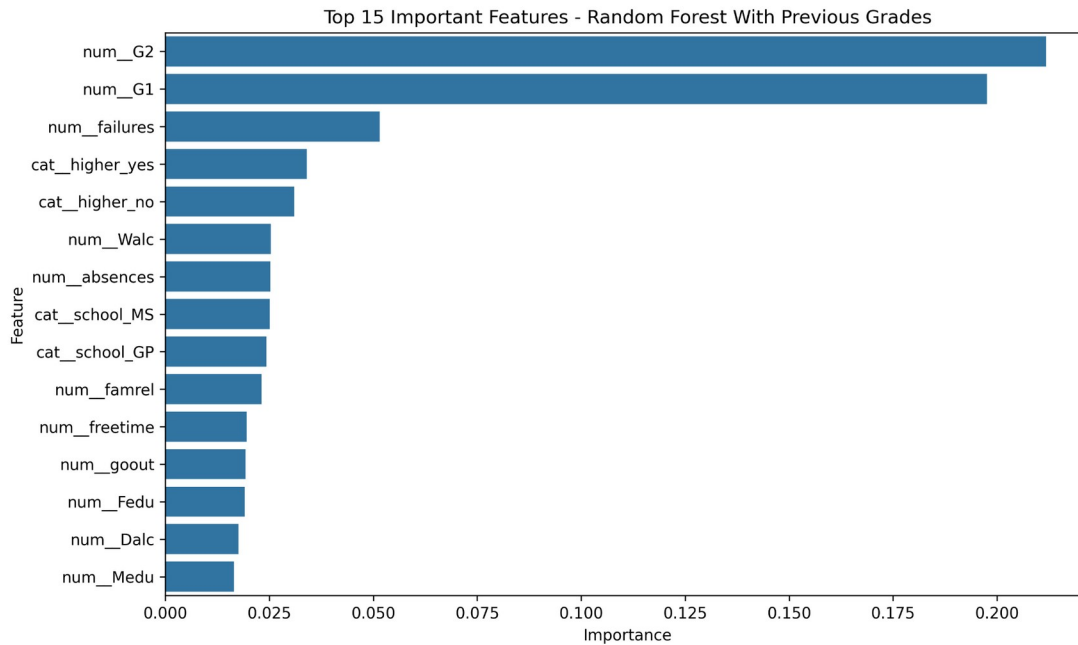
The best model, Logistic Regression with previous grades, correctly predicted 14 failing students and 104 passing students in the test set. It made 12 total errors: 6 failing students were predicted as passing, and 6 passing students were predicted as failing.

The strong performance of models that include G1 and G2 is expected because the first and second period grades are highly related to the final grade G3. This is also visible in the correlation chart and feature importance analysis.

The second experiment is valuable because it tests whether the model can predict student risk without previous grade information. Although performance decreases, SVM still achieved a strong F1-score of 0.9052. This suggests that non-grade factors can still provide useful early signals.

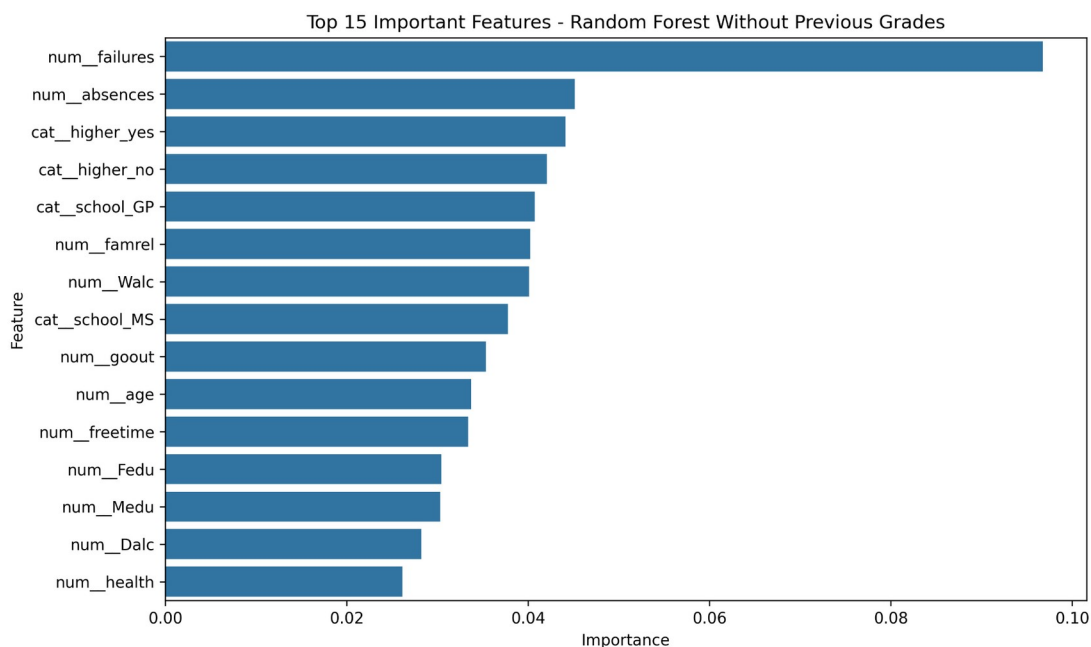
Feature Importance Analysis

Figure 8. Random Forest feature importance with previous grades



When G1 and G2 are included, G2 and G1 dominate the feature importance ranking. This confirms that previous academic performance is the strongest predictor of the final outcome.

Figure 9. Random Forest feature importance without previous grades



When G1 and G2 are excluded, the most influential variables include previous failures, absences, intention to pursue higher education, school, family relationship quality, weekend alcohol consumption, going out, and age. These features can help identify students who may need support before final grades are known.

9. Limitations

- Class imbalance exists because the Pass class is much larger than the Fail class. This makes accuracy less informative than F1-score and recall.
- The dataset comes from a specific educational context, so the model may not generalize directly to all schools or countries without validation on new data.
- Including G1 and G2 produces strong performance, but it may be less useful for very early prediction before grade records are available.
- Some features are sensitive social or behavioral variables. In a real school system, ethical data use, privacy, and fairness must be considered.
- The project used standard machine learning models. Hyperparameter tuning could further improve performance.

10. Conclusion and Future Work

The project successfully applied data mining techniques to predict student performance. It included data preprocessing, visualization, model training, model comparison, and interpretation of results.

The best overall model was Logistic Regression with previous grades, achieving accuracy of 0.9077 and F1-score of 0.9455. Random Forest also performed strongly with F1-score of 0.9417. Without previous grades, SVM produced the strongest early-warning result with F1-score of 0.9052.

The analysis shows that previous grades are the strongest predictors, but previous failures, absences, higher education intention, and other non-grade variables also provide useful signals. Future work could include hyperparameter tuning, cross-validation, fairness analysis, and deployment as an early-warning dashboard for academic advisors.

References

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Appendix: Reproducibility Notes

- Platform: Kaggle Notebook.
- Programming language: Python.
- Main libraries: pandas, numpy, matplotlib, seaborn, scikit-learn.
- Dataset file: student-por.csv.
- Target variable: pass_final, engineered from G3.
- Train/test split: 80/20, random_state = 42, stratified by target.
- Algorithms: Logistic Regression, Decision Tree, Random Forest, and SVM.
- Saved outputs: cleaned dataset, result table, visual charts, confusion matrix, and feature importance charts.